

REQUEST FOR INFORMATION (RFI)

Foundation Digital Twin Auto Feature Extraction (FDT AFE)

1.0 DESCRIPTION & PURPOSE

1.1 The National Geospatial-Intelligence Agency (NGA) is seeking information on acquiring an innovative solution to automate the identification, extraction, and attribution of geospatial features. The Automated Feature Extraction (AFE) capability will be a critical enabler for modernizing FG processes and will provide its capabilities as a service, accessible by key NGA systems, including FDT, via APIs.

1.2 The purpose of this Request for Information (RFI) is to gain information on current Industry and Academia practices and their capabilities that could meet NGA's requirements as described within the AFE Statement of Objectives (SOO). This RFI is focused on features which FG considers to be valuable but challenging.

1.3 THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for information and planning purposes – it does not constitute a Request for Proposal (RFP) or a promise to issue an RFP in the future. This RFI does not commit the Government to contract for any supply or service whatsoever. Further, NGA is not seeking proposals at this time and will not accept unsolicited proposals. Responders are advised that the U.S. Government will not pay for any information or administrative costs incurred in response to this RFI. All costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future RFP, if any is issued.

2.0 BACKGROUND

NGA has a mission-critical need for implementing the Automated Feature Extraction (AFE) to improve efficiency and effectiveness in mission data feature collection. FG Analysts spend extensive time on repetitive manual feature extraction, conflation, and validation, which are time-consuming activities and can potentially be automated. Implementing AFE as part of FG mission operations addresses these challenges and supports the NGA's 2035 GEOINT Concept of Operations. NGA envisions this initiative to leverage the use of artificial intelligence and machine learning (AI/ML) to transform geospatial intelligence production, significantly reducing manual processing time, increasing data accuracy, and enabling the rapid delivery of intelligence to decision-makers.

This RFI seeks industry input on available out-of-the-box AFE capabilities and how they can be provided as a service to support NGA operations to automatically detect and extract geospatial features from various data input sources, including imagery and raster maps, to meet the objectives described within the AFE Statement of Objectives (SOO), section 4.0.

2.1 PLANNED ACQUISITION: see attached SOO. This is a RFI which may develop into an RFP at a future date dependent upon responses received to this RFI.

2.2 PERFORMANCE: If a future RFP is released, it is envisioned to have the following period of performance (POP).

2.2.1 PERIOD (POP): 12-month base period, and four (4) 12-month option periods.

2.2.2 LOCATION(S): The primary place of performance for this effort shall be the contractor's facilities. The Government requires a solution that allows for the

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remote submission of processing jobs via Application Programming Interfaces (APIs) from Government systems to a contractor-provided platform. This platform must process the submitted jobs and return the resulting AFE data to the Government in the required format. Travel to Government facilities, such as NGA Washington (NGA-W) or NGA St. Louis (NGA-STL), shall be on an as-needed basis, as determined and approved by the Government.

2.3 CONSTRAINTS AND LIMITATIONS: The following are anticipated to apply to this acquisition:

- Solutions must contain a robust AFE capability that automatically detects and extracts geospatial features from various sources including imagery (EO, SAR, HS, MS) and raster maps. This system must be capable of monoscopic and/or stereoscopic extraction and handle individual and batch processing with varying levels of feature detail(s) over multiple geographical areas of interest.
- Capabilities shall automatically attribute extracted features based on observable characteristics in accordance with NGA domain standards and preserve feature geometry, spatial relationships, and metadata from the source data.
- Proposed solutions must preserve the source data's original geometry, spatial relationships, and metadata throughout the automated feature extraction process.
- Solutions should leverage existing interface solutions for analysts to efficiently review, validate, and correct extracted features.
- Solutions must provide confidence metrics to guide and prioritize data validation.
- Solutions must utilize standardized data formats, such as GeoDatabase (GDB), Shapefile, and TDS 7.1, to ensure compatibility with NGA systems.
- The solution must be fully compliant with all applicable security directives, including DNI ICD 503, and implement robust access controls based on user roles and clearances.
- Familiar with NGA policies for software development to include the NGA Data Strategy and the NGA Software Way.

2.4 SECURITY REQUIREMENTS: The following security requirements are anticipated to apply to this acquisition: *(Note that all of the personnel the responder anticipates performing the needed services would need to possess the appropriate security clearance at the time of proposal or quotation submission.)*

- Most of the work on this contract will be performed at the contractor site. Some contractor personnel shall possess a current TS/SCI clearance and be eligible for favorable NGA adjudication in order to access NGA W and NGA STL sites as determined by the Government. Contractors with a TS/SCI are subject to a Counterintelligence-Scope Polygraph examination as requested by NGA. The contractor must abide by the DD Form 254, the National Industrial Security Program Operating Manual (NISPOM), and all other applicable IC and NGA security policies and regulations, including those concerning information security, cybersecurity, and insider threat reporting.

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2.5 ORGANIZATIONAL CONFLICT OF INTEREST: This acquisition is for developmental service. In accordance with FAR Part 9.5, please discuss any performance which may lead to an OCI and how the Offeror would mitigate, avoid, or neutralize the conflict.

3.0 REQUESTED INFORMATION

Responses must only contain UNCLASSIFIED information and be marked “UNCLASSIFIED” on all pages in both the header and footer. No classified information may be included in your response.

Neither proprietary or classified concepts, nor classified information should be included in the submittal. Input on the information contained in the responses may be solicited and reviewed by NGA non-Government consultants or experts who are bound by appropriate non-disclosure agreements.

3.1 ADMINISTRATIVE

Information to include the following as a minimum:

3.1.1. Name, mailing address, overnight delivery address (if different from mailing address), phone number, fax number, company website, and e-mail of designated point(s) of contact.

3.1.2. BUSINESS TYPE.

Based upon [North American Industry Classification System \(NAICS\)](#) code 541370 - Surveying and Mapping (except Geophysical) Service applicable to this acquisition, the responder is to provide answers to the following questions:

<i>To be considered for a Small Business set-aside, at least 50 % of the cost of contract performance incurred for personnel shall be expended for employees of the responder. (FAR 52.219-14, Limitation of Subcontracting)</i>	RESPONSE
Are you a Small Business? (FAR 19.102), If so, which type:	☐ YES ☐ NO
Small Disadvantaged small business? (FAR 19.304)	☐ YES ☐ NO
Service Disabled small business? (FAR 19.14)	☐ YES ☐ NO
8(a) Small Disadvantaged small business? (FAR 19.8)	☐ YES ☐ NO
HUBZone small business? (FAR 19.13)	☐ YES ☐ NO
Woman-Owned Small Business? (FAR 19.15)	☐ YES ☐ NO
Involved in a mentor and/or protégé program? (DFARS 219.71)	☐ YES ☐ NO

3.1.3 Recommendation for alternative NAICS Code? Provide rationale for any NAICS Code applicable to the planned acquisition other than the NAICS code identified above.

3.1.4 DUNS Number: _____

3.1.5 Commercial and Government Entity (CAGE) Code: _____

3.1.6 Defense facility security clearance? What type? What level? _____

3.1.7 Accounting system, *if applicable*? Date of last audit? Audit performed by? Determined adequate/inadequate? _____

3.1.8 Purchasing system? Date determined adequate or inadequate? _____

3.1.9 Timekeeping system? Date of last audit? Who performed? Determined adequate/inadequate? _____

3.1.10 Do you hold a GSA Federal Supply Schedule (FSS) contract applicable to the NAICS code (s) identified above? If yes, provide contract number(s). _____

3.1.11 Are you registered in SAM? _____

3.1.12 Are you interested in: Prime contract? Teaming to include Joint Venture? Subcontractor opportunities? _____

3.1.13 Include additional details that are not already requested based on the constraints of the information request.

3.2 EXPERIENCE

3.2.1 COMMERCIAL.

Provide relevant details on the responder providing the same or similar services offered or made to the general public or to non-governmental entities for purposes other than governmental purposes in the last three (3) years. Relevant details to NGA's proposed acquisition should include, but not be limited to, information regarding the contract value, size and length of the effort, responder performing as a prime or subcontractor, customary practices (warranty, financing, discounts, contract types, etc.) under which the sales of the service(s) are made, security details, the customary practices regarding customizing, modifying, or tailoring of a service(s) to meet customer needs and associated costs, the kinds of factors that are used to evaluate performance; the kinds of performance incentives used; the kinds of performance assessment methods commonly used; the common qualifications of the people performing the services; and requirements of law and/or regulations unique to these service(s), that can demonstrate the responder's abilities and capacity to meet NGA's statement of objectives.

3.2.2 GOVERNMENT.

Provide relevant details on the responder providing the same or similar services offered or made to Government agencies other than NGA in the last three (3) years. Relevant details to NGA's proposed acquisition should include, but not be limited to contract value, information regarding the contract number, agency, responder performing as a prime or subcontractor, size and length of the effort, type of pricing and/or cost, security details, the kinds of factors used to evaluate performance; the kinds of performance incentives; the kinds of performance assessment methods; the qualifications of the people performing the services; and any unique terms and conditions, that can demonstrate the responder's abilities and capacity to meet NGA's statement of objectives.

3.2.3 NGA.

Provide relevant details on the responder providing the same or similar services offered or made to NGA in the last three (3) years. Relevant details to NGA's proposed acquisition should

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include, but not be limited to contract value, information regarding the contract number, program name, size and length of the effort, responder performing as a prime or subcontractor, type of pricing and/or cost, security details, the kinds of factors used to evaluate performance; the kinds of performance incentives; the kinds of performance assessment methods; the qualifications of the people performing the services; and any unique terms and conditions, that can demonstrate the responder's abilities and capacity to meet NGA's statement of objectives.

3.3 CAPABILITIES

The following is the responder's capabilities to meet the RFI needs set forth in Attachment 3 - Automated Feature Extraction RFI Objective Questions and the AFE SOO, dated [February 25, 2026](#).

3.3.1 Government requirements needed to support the RFI FDT AFE SOO:

3.3.1.1 Requirement capabilities that the responder currently possesses. Responses should include the relevant information regarding specific skills, experience, and security clearances that its employees, by labor category, currently possess in performing these services, and any needed hardware or software.

3.3.1.2 Detail the service capabilities that the responder currently does not possess in order to meet the SOO.

3.3.2 Recommended type of pricing for this type of service (e.g. FFP, FPIF CPIF).

3.3.3 Extent to which the responder has the ability to meet a future RFP.

3.3.4 Notional schedule and type of plan for transition-in and transition-out (based on previous or similar work efforts).

3.3.5 Security capabilities and plans that demonstrate the ability to meet NGA's security requirements beginning at contract award and throughout the POP.

3.3.6 Extent to which the responder is aware of any potential OCI issues in accordance with FAR Part 9.5 or any non-mitigatable OCI related to current development work at NGA.

3.3.7 The response should specifically detail the solution's proficiency across the three primary AFE functions: (A) Object Detection, (B) Geometry Extraction, and (C) Attribution, as defined in Attachment 4 – Response to USE CASES.

3.3.8 The requested capability must support multiple file formats (including EO, SAR, HS, and MS) and raster maps. It shall be capable of both monoscopic and stereoscopic feature extraction, and support both individual and batch processing across multiple geographic areas of interest with varying levels of feature detail.

3.5 RECOMMENDATIONS

The responder is invited to provide information and recommendations for fashioning this proposed acquisition. Recommendation areas may include the type of contract, anticipated contract terms & conditions, incentives, variations in delivery schedule, price and/or cost proposal support, and data requirements, contract pricing, and any other areas that the responder believes are relevant for the Government to achieve its stated objectives. The responder may opine regarding what portion of the work could be set aside for small businesses if all the

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required work cannot be provided by a small business. Details to support any recommendations for partial small business set-asides should be included.

4.0 RESPONSES

4.1 Interested parties are directed to respond to this RFI with direct responses to each question outlined in this RFI document as well as any relevant information about the company that is at an unclassified level. ***Maximum page count is three (3) for individual use cases and nine (9) for RFI response to questions for a total of twelve (9) + (3 X # use cases) pages.**

*For example: Respondent A responds to 3 of the 6 use cases and then provides their RFI response. Total pages would be 3 uses cases times 3 pages per use case plus 9 pages for RFI response for a total of 18 pages.

4.2 Responses shall be in Microsoft Word for Office or compatible format and shall not exceed 27 pages total (this page total assumes a response to all use cases). A page is defined as each face of an 8 ½" x 11" sheet with information contained within a one-inch margin on all sides. Font type shall be Times Roman 12 point.

4.3 Interest parties are encouraged to utilize the comment Matrix format in their response when proving comments, questions, and/or suggestions to attached document. Filling out the comment Matrix is optional and does not count toward the overall page limit.

4.4 Responses are due no later than 5:00 pm ET on **06 April 2026**. Responses shall be limited to and submitted via e-mail only as a message attachment to Dan Fadely at Daniel.R.Fadely@nga.mil and Dee Hill at Delores.M.Hill@nga.mil with the message subject line "RFI Response – Foundation Digital Twin Automated Feature Extraction

4.5 Proprietary information, if any, should be minimized and **MUST BE CLEARLY MARKED**. To aid the Government, please segregate proprietary information. Please be advised that all submissions become Government property and will not be returned.

5.0 MEETINGS AND DISCUSSIONS

The Government representatives may or may not choose to meet with potential RFI service provider. Such meetings and discussions would only be intended to get further clarification of potential capability, especially any development and certification risks.

6.0 QUESTIONS

Questions and the whitepaper regarding this RFI should be submitted in writing by e-mail to Dan Fadely at Daniel.R.Fadely@nga.mil and Dee Hill at Delores.M.Hill@nga.mil.

7.0 SUMMARY

This RFI is only intended for the Government to identify sources that can provide the services needed to fulfill the Government's SOO. The information provided in this RFI is subject to change and is not binding to the Government. The Government has not made a commitment to procure any of the RFI requirements discussed, and release of this RFI should not be construed as such a commitment or as authorization to incur cost for which reimbursement would be required or sought. All submissions become Government property and will not be returned.

*Foundation Digital Twin Automated Feature Extraction***STATEMENT OF OBJECTIVES (SOO)****COMMENT MATRIX FOR THE FOUNDATION DIGITAL TWIN AUTOMATED FEATURE
EXTRACTION**

COMMENT #	SOO PAGE #	SOO /SECTION	COMMENT/QUESTION/SUGGESTION
1			
2			
3			
4			
5			

Attachment 1: Statement of Objective dated 25 Feb 2026

Attachment 2: Request for Information dated March 5, 2026

Attachment 3: Automated Feature Extraction RFI Objective Questions

Attachment 4: Responses to USE CASES

Appendix A: NGA Data Strategy

Appendix B: NGA Software Way